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## Software Requirements Specification

|       |       |
|-------|-------|
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## 1. Introduction

### 1.1 Purpose

This document specifies the functional and non-functional requirements for NOVA, a locally hosted agent that converts software requirements into structured test plans and retains user corrections as durable memory. It is the authoritative requirements baseline for the initial release and the source against which verification is performed.

Intended audience: the implementing engineer, and any reviewer assessing the completeness or testability of the requirement set.

### 1.2 Scope

The software product is named **NOVA**.

NOVA accepts a natural-language software requirement, classifies it by type, and produces a structured test plan conforming to a versioned schema. It records user corrections at test case granularity, extracts durable rules from those corrections subject to explicit user confirmation, and applies them to subsequent structurally similar requirements. It exposes a complete execution trace for every task and measures whether corrections recur.

NOVA does not execute tests, generate executable test code, ingest source repositories, or integrate with issue trackers. It performs no model training of any kind.

Benefits and objectives are stated in [NOVA-VS-001](#).

### 1.3 Product overview

#### 1.3.1 Product perspective

NOVA is a self-contained product with no predecessor system. It consists of four deployable components: a browser-based user interface, a REST API service, an asynchronous worker process, and a relational datastore with vector indexing. Inference is supplied by an external runtime resident on the host, addressed through an internal provider abstraction.

External interfaces are limited to the inference runtime. The system has no other outbound network dependency in its default configuration.

### 1.3.2 Product functions

| Function             | Summary                                                                              |
|----------------------|--------------------------------------------------------------------------------------|
| Requirement intake   | Accept, validate, classify, and persist a submitted requirement                      |
| Test plan generation | Produce a schema-conformant test plan using constrained decoding                     |
| Memory retrieval     | Select applicable stored rules and expose the selection scores                       |
| Strategy selection   | Choose a generation playbook from recorded outcome scores                            |
| Quality evaluation   | Apply deterministic checks and a bounded model-based review before returning results |
| Feedback capture     | Record accept, edit, reject, and addition events at test case granularity            |
| Memory extraction    | Derive candidate rules from corrections and present them for confirmation            |
| Memory management    | Browse, edit, pin, and delete stored rules                                           |
| Execution tracing    | Persist and present a complete record of each task execution                         |
| Evaluation           | Execute a versioned dataset, compute metrics, and gate configuration changes         |

### 1.3.3 User characteristics

| Class        | Description                                                                                                                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary user | Software or test engineer, two to six years of experience, fluent in test design and comfortable with command-line tooling and HTTP APIs. Holds specific conventions regarding case format, mandatory case types, and priority taxonomy. Professionally disinclined to accept unexplained automated output |
| Operator     | The same individual. Responsible for deployment, model provisioning, and backup                                                                                                                                                                                                                            |

There is no administrative user class in the initial release.

### 1.3.4 Limitations

| ID    | Limitation                                                   | Origin                |
|-------|--------------------------------------------------------------|-----------------------|
| CON-1 | Development effort limited to approximately 130 to 180 hours | Resource availability |
| CON-2 | Single engineer, no external code review                     | Project structure     |
| CON-3 | Inference constrained to a 6 GB VRAM budget                  | Measured hardware     |
| CON-4 | Deployment is local only                                     | Project decision      |
| CON-5 | Synthetic and open-source data only                          | Project decision      |
| CON-6 | No model fine-tuning                                         | Scope boundary        |
| CON-7 | Single user, with a multi-tenant-capable schema              | Project decision      |

## 1.4 Definitions, acronyms, and abbreviations

| Term                 | Definition                                                                                                        |
|----------------------|-------------------------------------------------------------------------------------------------------------------|
| AC                   | Acceptance criterion. A verifiable condition stated within a requirement                                          |
| ADR                  | Architecture Decision Record                                                                                      |
| Applied lesson       | A stored memory retrieved and injected into a generation prompt for a given task                                  |
| Candidate memory     | An extracted rule awaiting user confirmation. Not yet retrievable                                                 |
| Constrained decoding | Generation restricted at the token level to output conforming to a supplied schema                                |
| CRR                  | Correction Recurrence Rate. See section 3.1.9                                                                     |
| Execution trace      | The ordered, persisted record of every step, model call, and score for one task                                   |
| Memory               | A durable rule derived from a user correction, with provenance, scope, and confidence                             |
| Playbook             | A named, versioned generation strategy comprising required case types, a prompt fragment, and check configuration |
| Provenance           | The originating task, correction, and timestamp recorded against a memory                                         |
| Tenant               | The isolation boundary for all user-owned data                                                                    |
| Test plan            | The structured output of one generation, comprising a summary and an ordered set of test cases                    |
| Tombstone            | The residual record of a deleted memory, retaining identifiers and timestamps but no content                      |

## 2. References

| Reference                 | Title                                                                            |
|---------------------------|----------------------------------------------------------------------------------|
| NOVA-VS-001               | Vision and Scope                                                                 |
| NOVA-SAD-001              | Software Architecture Document                                                   |
| NOVA-STP-001              | Software Test Plan                                                               |
| NOVA-TM-001               | Threat Model                                                                     |
| ISO/IEC/IEEE 29148:2018   | Systems and software engineering, life cycle processes, requirements engineering |
| ISO/IEC/IEEE 29119-3:2021 | Software testing, test documentation                                             |
| RFC 9457                  | Problem Details for HTTP APIs                                                    |
| arc42 v9                  | Architecture documentation template                                              |

### 3. Specific requirements

Requirements use `shall` for mandatory behaviour. Each is uniquely identified, individually verifiable, and traced in section 4.

Priority values follow MoSCoW: **M** must, **S** should, **C** could, **W** will not.

#### 3.1 Functions

##### 3.1.1 Requirement intake

| ID     | Requirement                                                                                                                                                       | Pri |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-1   | The system shall accept a requirement consisting of natural-language text, a requirement type, and optional supplementary context                                 | M   |
| FR-1.1 | The system shall reject a requirement whose text is empty or exceeds the configured length limit, returning a message identifying the violated constraint         | M   |
| FR-1.2 | The system shall persist every accepted requirement together with its submission timestamp and the identifier of the configuration version in force at submission | M   |

##### 3.1.2 Test plan generation

| ID     | Requirement                                                                                                                                                | Pri |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-2   | The system shall generate a test plan conforming to the versioned test plan schema, using constrained decoding to enforce structural validity              | M   |
| FR-2.1 | The system shall record the schema version against every generated plan and shall remain able to read plans produced under earlier schema versions         | M   |
| FR-2.2 | Where a generated plan fails schema validation, the system shall attempt a bounded number of repair retries before recording the task as failed            | M   |
| FR-2.3 | The system shall record every generation attempt, whether successful or not, against the validity metric                                                   | M   |
| FR-2.4 | Where the assembled prompt would exceed the available context budget, the system shall reduce retrieved memory content before reducing requirement content | M   |

##### 3.1.3 Strategy selection

| ID     | Requirement                                                                                                                         | Pri |
|--------|-------------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-3   | The system shall select one playbook per task from at least two available playbooks, using recorded outcome scores                  | M   |
| FR-3.1 | The system shall apply bounded random exploration at a configurable rate with a non-zero lower bound                                | M   |
| FR-3.2 | The system shall apply a static default playbook where the observation count for a requirement type is below the configured minimum | M   |
| FR-3.3 | The system shall constrain each outcome score update to a bounded step and shall clamp scores to the interval [-1, +1]              | M   |

| ID     | Requirement                                                                                                          | Pri |
|--------|----------------------------------------------------------------------------------------------------------------------|-----|
| FR-3.4 | The system shall record whether a given selection was exploratory, and shall present that fact in the user interface | M   |
| FR-3.5 | The system shall be able to recompute all outcome scores from the recorded feedback event history                    | M   |

#### *3.1.4 Memory retrieval*

| ID     | Requirement                                                                                                                                            | Pri |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-4   | The system shall retrieve stored memories ranked by a weighted combination of semantic similarity, recency, and confidence                             | M   |
| FR-4.1 | The system shall apply requirement type and tenant as filters prior to ranking                                                                         | M   |
| FR-4.2 | The system shall return memories marked as pinned and in scope irrespective of their rank                                                              | M   |
| FR-4.3 | The system shall exclude memories scoring below the configured relevance threshold, and shall record the exclusion                                     | M   |
| FR-4.4 | The system shall present each applied memory to the user together with its component and combined scores                                               | M   |
| FR-4.5 | Where retrieval returns no memory above threshold, the system shall proceed using playbook defaults and shall state that no prior lesson was applied   | M   |
| FR-4.6 | Where the embedding service is unavailable, the system shall fail the task with an explicit cause and shall not proceed with an empty retrieval result | M   |

#### *3.1.5 Feedback capture*

| ID     | Requirement                                                                                                           | Pri |
|--------|-----------------------------------------------------------------------------------------------------------------------|-----|
| FR-5   | The system shall record accept, edit, reject, and addition events against individual test cases                       | M   |
| FR-5.1 | The system shall accept an optional freeform correction against a task                                                | M   |
| FR-5.2 | The system shall retain the originally generated text of an edited case alongside the edited text                     | M   |
| FR-5.3 | The system shall normalize each feedback event to a score in the interval [-1, +1] using a versioned weighting scheme | M   |
| FR-5.4 | The system shall treat feedback events as immutable once written                                                      | M   |

#### *3.1.6 Memory extraction and confirmation*

| ID   | Requirement                                                       | Pri |
|------|-------------------------------------------------------------------|-----|
| FR-6 | The system shall derive candidate rules from freeform corrections | M   |

| ID     | Requirement                                                                                                                                                                                      | Pri |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-6.1 | The system shall reject a candidate that is not generalizable, is not actionable, claims a scope broader than its evidence, exceeds the configured length limit, or is empty of semantic content | M   |
| FR-6.2 | The system shall flag for review any candidate whose text is shaped as an instruction directed at the system, and shall not store such a candidate without explicit user action                  | M   |
| FR-6.3 | The system shall not store any memory without explicit user confirmation                                                                                                                         | M   |
| FR-6.4 | The system shall detect near-duplicate candidates and shall reinforce the existing memory rather than creating a second record                                                                   | M   |
| FR-6.5 | The system shall detect a candidate that contradicts an existing in-scope memory and shall present both to the user for resolution                                                               | M   |
| FR-6.6 | Where extraction fails, the system shall retain the original correction text for later processing                                                                                                | M   |
| FR-6.7 | The system shall record provenance against every stored memory, identifying the originating task, correction, and timestamp                                                                      | M   |

#### *3.1.7 Memory management*

| ID     | Requirement                                                                                                                                                           | Pri |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-7   | The system shall allow the user to browse stored memories, filtered by scope, status, and pinned state                                                                | M   |
| FR-7.1 | The system shall allow the user to edit the text, scope, and pinned state of a stored memory                                                                          | M   |
| FR-7.2 | The system shall re-derive the embedding and reset confidence to its initial value when memory text is edited                                                         | M   |
| FR-7.3 | The system shall allow the user to delete a stored memory, removing its content and rendering it unretrievable                                                        | M   |
| FR-7.4 | The system shall retain the identifier, timestamps, and tenant of a deleted memory so that historical traces remain coherent                                          | M   |
| FR-7.5 | The system shall reduce the confidence of a memory that remains unapplied over time, and shall archive rather than delete a memory falling below the configured floor | M   |
| FR-7.6 | The system shall link a superseding memory to the memory it replaces and shall retain the superseded record                                                           | M   |
| FR-7.7 | The system shall export the complete memory set in a machine-readable format                                                                                          | S   |

#### *3.1.8 Execution tracing*

| ID | Requirement | Pri |
|----|-------------|-----|
|----|-------------|-----|

| ID     | Requirement                                                                                                                                                                                             | Pri |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-8   | The system shall persist an execution trace for every task, comprising the playbook selection and rationale, all retrieval results with scores, every model call, and every deterministic check verdict | M   |
| FR-8.1 | The system shall record retrieval scores for candidate memories that were evaluated but not selected                                                                                                    | M   |
| FR-8.2 | The system shall record prompt token count, completion token count, latency, attempt number, and validity outcome for each model call                                                                   | M   |
| FR-8.3 | The system shall write trace steps as they complete, such that a task failing part-way retains the trace of all completed steps                                                                         | M   |
| FR-8.4 | The system shall store a cryptographic digest of each prompt in place of the prompt text                                                                                                                | M   |

#### *3.1.9 Evaluation and configuration control*

| ID      | Requirement                                                                                                                                                               | Pri |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-9    | The system shall compute and persist the metrics defined in NOVA-STP-001 for each task and each evaluation run                                                            | M   |
| FR-9.1  | The system shall execute an evaluation suite against a content-hash-versioned dataset and report per-metric aggregates                                                    | M   |
| FR-9.2  | The system shall compute the Correction Recurrence Rate as the proportion of previously confirmed corrections that reappear as errors on later structurally similar tasks | M   |
| FR-10   | The system shall maintain versioned prompt and configuration records                                                                                                      | M   |
| FR-10.1 | The system shall reject activation of a configuration version that is not linked to a passing evaluation run, and shall enforce this at the data layer                    | M   |
| FR-10.2 | The system shall retain superseded configuration versions                                                                                                                 | M   |

#### *3.1.10 Inference provider management*

| ID      | Requirement                                                                                                                   | Pri |
|---------|-------------------------------------------------------------------------------------------------------------------------------|-----|
| FR-11   | The system shall support multiple inference providers behind a common interface, defaulting to local execution                | M   |
| FR-11.1 | The system shall change the active provider only on explicit user action, and shall never substitute a provider automatically | M   |
| FR-11.2 | The system shall display the active provider in the user interface                                                            | M   |
| FR-11.3 | The system shall report inference provider reachability through its readiness endpoint                                        | M   |
| FR-     | Where a required model is absent, the system shall return a message naming the model                                          | M   |



| ID   | Requirement                          | Pri |
|------|--------------------------------------|-----|
| 11.4 | and the action required to obtain it |     |

### 3.1.11 Presentation

| ID    | Requirement                                                                                   | Pri |
|-------|-----------------------------------------------------------------------------------------------|-----|
| FR-12 | The system shall display accumulated token count and elapsed duration for each completed task | S   |
| FR-13 | The system shall export a test plan in Markdown and comma-separated value formats             | C   |
| FR-14 | The system shall present two executions of the same requirement in a comparison view          | C   |
| FR-15 | The system shall support memory sharing between users                                         | W   |

## 3.2 Performance requirements

Targets marked *provisional* are set from judgment and are subject to revision once the corresponding measurement exists. Where a measurement exists it is cited.

| ID    | Requirement                                              | Target  | Basis                                                                                                      |
|-------|----------------------------------------------------------|---------|------------------------------------------------------------------------------------------------------------|
| NFR-1 | Structured output shall be schema-valid on first attempt | ≥98%    | Provisional. 12 of 12 observed over a 12-generation sample. See <a href="#">spike</a>                      |
| NFR-2 | End-to-end task duration, 95th percentile                | ≤90 s   | Provisional and known to be slack. Generation alone measured 9 to 26 s warm on the reference configuration |
| NFR-3 | Memory retrieval duration, 95th percentile               | ≤200 ms | Provisional                                                                                                |
| NFR-4 | Evaluation suite total duration                          | ≤15 min | Provisional. Cold model load is a material contributor and shall be reported separately                    |
| NFR-5 | Inference provider call failure rate                     | ≤2%     | Provisional                                                                                                |

Reference configuration for all timing requirements: NVIDIA RTX 4050 Laptop, 6 GB VRAM, 4B-parameter model at 4-bit quantization, 8192-token context. Timing requirements are void without this qualification.

## 3.3 Usability requirements

| ID    | Requirement                                                                         | Pri |
|-------|-------------------------------------------------------------------------------------|-----|
| NFR-6 | The user interface shall remain responsive to interaction while a task is executing | M   |

| ID    | Requirement                                                                                                                      | Pri |
|-------|----------------------------------------------------------------------------------------------------------------------------------|-----|
| NFR-7 | The system shall present task status such that a queued, running, failed, or cancelled task is distinguishable without ambiguity | M   |
| NFR-8 | The system shall permit cancellation of a running task, retaining the partial trace                                              | M   |
| NFR-9 | Error messages presented to the user shall identify the failing step and the corrective action where one exists                  | M   |

### 3.4 Interface requirements

#### 3.4.1 User interfaces

Four views: task submission and results, execution trace, memory management, and evaluation results. Rendered in a current desktop browser. No mobile layout is provided.

The screen inventory, wireframes, state coverage, and element-to-requirement traceability are in [NOVA-UI-001](#).

#### 3.4.2 Application programming interfaces

| ID     | Requirement                                                                                                                                                                                | Pri |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| NFR-10 | The system shall expose a versioned HTTP interface under a path prefix, admitting only additive change within a version                                                                    | M   |
| NFR-11 | The system shall return errors in the format defined by RFC 9457                                                                                                                           | M   |
| NFR-12 | The system shall honour an idempotency key on task creation, returning the original task identifier on replay                                                                              | M   |
| NFR-13 | The system shall paginate collection responses using opaque cursors                                                                                                                        | M   |
| NFR-14 | The system shall publish an interface description generated from its request and response models, and shall fail its build where the published description diverges from the generated one | M   |

#### 3.4.3 External interfaces

The inference runtime is addressed over HTTP on the host loopback interface. The system supplies a JSON Schema with each generation request and receives a schema-conformant completion together with token and timing telemetry.

### 3.5 System operations

#### 3.5.1 Nominal operation

1. The user submits a requirement with its type.

2. The system validates the submission, persists it, enqueues a job, and returns a task identifier without waiting for execution.
3. A worker claims the job, reads the active configuration version, selects a playbook, retrieves applicable memories, generates a plan under schema constraint, applies deterministic checks and a bounded model-based review, and persists the plan and trace.
4. The user reviews the plan alongside the applied lessons and their scores.
5. The user records accept, edit, or reject against individual cases, and may supply a freeform correction.
6. The system derives candidate rules and presents them for confirmation.
7. On confirmation, the system embeds and stores each rule.
8. On a subsequent structurally similar requirement, the stored rules are retrieved and applied, and are shown as applied lessons.

### 3.5.2 Exception operation

| ID    | Condition                                                       | Required behaviour                                                                                |
|-------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| OP-1  | Memory store empty                                              | Generate from playbook defaults. State explicitly that no prior lesson was applied                |
| OP-2  | Generated output fails schema validation                        | Bounded repair retry, then terminal failure with a stated cause. All attempts recorded            |
| OP-3  | Inference provider unreachable                                  | Terminal failure with a stated cause. Previously completed results remain accessible              |
| OP-4  | Retrieved memories conflict                                     | Present the conflict. Do not resolve it silently                                                  |
| OP-5  | Requirement text contains an instruction directed at the system | Treat as data. Neutralize and record                                                              |
| OP-6  | All cases rejected                                              | Record as a strongly negative outcome. Prompt for a freeform correction                           |
| OP-7  | Retrieval returns nothing above threshold                       | Proceed on defaults. Record the empty result                                                      |
| OP-8  | A memory in current use is deleted                              | Honour the deletion immediately and completely. Retain a tombstone reference in historical traces |
| OP-9  | Task exceeds its duration budget                                | Remain cancellable. Preserve the partial trace                                                    |
| OP-10 | Later feedback contradicts an earlier correction                | Apply recency and confidence weighting. Supersede the earlier rule. Do not average                |
| OP-11 | Worker process terminates during execution                      | Reclaim the job after lease expiry. Preserve the partial trace                                    |
| OP-12 | Datastore unreachable                                           | Fail fast. Perform no partial writes                                                              |

### 3.6 System modes and states

Task states and their permitted transitions:

| State     | Permitted successors         |
|-----------|------------------------------|
| queued    | running, cancelled           |
| running   | completed, failed, cancelled |
| completed | terminal                     |
| failed    | terminal                     |
| cancelled | terminal                     |

Memory states:

| State      | Permitted successors          |
|------------|-------------------------------|
| candidate  | active, discarded             |
| active     | superseded, archived, deleted |
| archived   | active, deleted               |
| superseded | deleted                       |
| deleted    | terminal                      |

Configuration version states: draft, evaluated, active, retired. Transition to active requires an associated passing evaluation run.

### 3.7 Physical characteristics

Not applicable. NOVA is a software-only product with no hardware deliverable.

### 3.8 Environmental conditions

The system operates on a single workstation running Windows 11 with a container runtime and a GPU-accelerated inference runtime resident on the host. No requirement is placed on temperature, humidity, shock, or electromagnetic environment.

### 3.9 System security

| ID     | Requirement                                                                                                       | Pri |
|--------|-------------------------------------------------------------------------------------------------------------------|-----|
| NFR-15 | The system shall associate every user-owned record with a tenant identifier                                       | M   |
| NFR-16 | The system shall apply tenant scoping at the data access layer such that no query path can omit it                | M   |
| NFR-17 | The system shall resolve the tenant identifier server-side and shall never accept it from a client-supplied value | M   |

| ID     | Requirement                                                                                                                                                                         | Pri |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| NFR-18 | The system shall permit zero cross-tenant data disclosure. This requirement admits no tolerance                                                                                     | M   |
| NFR-19 | The system shall treat requirement text, correction text, and model output as untrusted data, delimited and labelled within prompts and never concatenated into instruction context | M   |
| NFR-20 | The system shall provide the inference model with no capability to read files, execute commands, or initiate network requests                                                       | M   |
| NFR-21 | The system shall validate all model output against its schema before persistence or presentation                                                                                    | M   |
| NFR-22 | The system shall escape untrusted content on presentation                                                                                                                           | M   |
| NFR-23 | The system shall exclude requirement text, correction text, and memory content from application logs by default                                                                     | M   |
| NFR-24 | The system shall bound every retry loop and shall define a terminal state for each                                                                                                  | M   |
| NFR-25 | The system shall hold no credential in its default configuration                                                                                                                    | M   |

Threat analysis is in [NOVA-TM-001](#).

### 3.10 Information management

| ID     | Requirement                                                                                                                  | Pri |
|--------|------------------------------------------------------------------------------------------------------------------------------|-----|
| NFR-26 | The system shall treat feedback events as the authoritative record and shall derive all outcome scores from them             | M   |
| NFR-27 | The system shall retain feedback events independently of the lifecycle of any memory derived from them                       | M   |
| NFR-28 | The system shall version feedback normalization weights such that historical scores can be recomputed under a revised scheme | M   |
| NFR-29 | The system shall treat execution traces as append-only and immutable once written                                            | M   |
| NFR-30 | The system shall version the evaluation dataset by content hash and shall refuse to compare runs across dataset versions     | M   |
| NFR-31 | The system shall support a documented backup and restore procedure                                                           | M   |

### 3.11 Policies and regulations

No regulatory regime applies. The system processes synthetic and open-source data only and transmits no data beyond the host in its default configuration. Personally identifiable information detection is not implemented; this is a recorded gap, and its status changes should authentic data enter the system.

### 3.12 System life cycle sustainment

| ID     | Requirement                                                                                                              | Pri |
|--------|--------------------------------------------------------------------------------------------------------------------------|-----|
| NFR-32 | The system shall start from a clean checkout by a single documented command                                              | M   |
| NFR-33 | The system shall apply schema changes through forward-only versioned migrations                                          | M   |
| NFR-34 | The system shall reproduce a prior execution given the same input, random seed, model version, and configuration version | M   |
| NFR-35 | Automated tests covering memory, retrieval, scoring, and validation logic shall achieve at least 80% statement coverage  | M   |

### 3.13 Packaging, handling, shipping, and transportation

Not applicable.

## 4. Verification

Each requirement is verified by one of: **T** automated test, **D** demonstration, **A** analysis or inspection, **M** measurement against the evaluation dataset.

| Requirement group | Method  | Verification artifact                                                                                                                    |
|-------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1 to FR-2.4    | T, M    | Schema conformance tests; structured output validity metric                                                                              |
| FR-3 to FR-3.5    | T, D    | Seeded selection tests; score recomputation test; poisoning scenario                                                                     |
| FR-4 to FR-4.6    | T, M    | Ranking unit tests; retrieval precision against a labelled relevance set; cross-tenant isolation suite                                   |
| FR-5 to FR-5.4    | T       | Normalization tests across all event types; immutability test                                                                            |
| FR-6 to FR-6.7    | T, A, M | Validation rejection tests; adversarial injection-into-memory scenarios; extraction precision measured against hand-labelled corrections |
| FR-7 to FR-7.7    | T, D    | Deletion completeness test; tombstone audit; lifecycle transition tests                                                                  |
| FR-8 to FR-8.4    | T       | Trace completeness test on a deliberately failed task; log content assertion                                                             |
| FR-9 to FR-10.2   | T, M    | Harness self-test; regression gate demonstration on a deliberately degraded configuration                                                |
| FR-11 to FR-      | T, D    | Provider contract suite; explicit switch test; readiness endpoint test                                                                   |

| Requirement group | Method | Verification artifact                                                                                                    |
|-------------------|--------|--------------------------------------------------------------------------------------------------------------------------|
| 11.4              |        |                                                                                                                          |
| NFR-1 to NFR-5    | M      | Instrumented measurement on the stated reference configuration                                                           |
| NFR-6 to NFR-9    | D      | Demonstration and exploratory session                                                                                    |
| NFR-10 to NFR-14  | T      | Interface description drift test; idempotency test; pagination tests                                                     |
| NFR-15 to NFR-25  | T, A   | Cross-tenant isolation suite as an unconditional gate; adversarial suite; log content assertion; architecture inspection |
| NFR-26 to NFR-31  | T, D   | Score recomputation test; restore drill                                                                                  |
| NFR-32 to NFR-35  | D, T   | Clean-checkout deployment on a separate host; reproducibility check; coverage gate                                       |

Acceptance is complete when every requirement of priority **M** has passed its stated verification method and the results are recorded.

## Appendix A. Assumptions and dependencies

| ID    | Assumption                                                                                          | Status                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| ASM-1 | Local inference sustains schema fidelity at acceptable latency                                      | Confirmed. 12 of 12 valid, 9 to 26 s warm, fully GPU-resident                                  |
| ASM-2 | A local model extracts reusable rules from freeform corrections at acceptable precision             | Unverified. Highest project uncertainty. Fallback is user-authored rules with model assistance |
| ASM-3 | Structural similarity between requirements is definable with sufficient stability to support FR-9.2 | Unverified                                                                                     |
| ASM-4 | Deterministic checks capture sufficient quality signal to serve as a gate                           | Unverified                                                                                     |
| ASM-5 | Inference nondeterminism remains within evaluation gate margins under a fixed seed                  | Unverified                                                                                     |
| ID    | Dependency                                                                                          | Status                                                                                         |
| DEP-1 | Inference runtime, host-resident                                                                    | Installed                                                                                      |
| DEP-2 | Model weights                                                                                       | gemma3:4b for generation, embeddinggemma for embedding. Both selected by measurement           |

| ID    | Dependency                                             | Status                                              |
|-------|--------------------------------------------------------|-----------------------------------------------------|
| DEP-3 | Relational datastore with vector indexing              | Not provisioned                                     |
| DEP-4 | Container runtime                                      | Installed, not running                              |
| DEP-5 | Embedding model, and consequently the vector dimension | embeddinggemma, 768 dimensions. Selected 2026-09-08 |

## Appendix B. Acronyms and abbreviations

| Abbreviation | Expansion                           |
|--------------|-------------------------------------|
| AC           | Acceptance criterion                |
| ADR          | Architecture Decision Record        |
| API          | Application Programming Interface   |
| CI           | Continuous Integration              |
| CRR          | Correction Recurrence Rate          |
| CSV          | Comma-Separated Values              |
| FR           | Functional Requirement              |
| NFR          | Non-Functional Requirement          |
| MoSCoW       | Must, Should, Could, Will not       |
| REST         | Representational State Transfer     |
| SAD          | Software Architecture Document      |
| SRS          | Software Requirements Specification |
| VRAM         | Video Random Access Memory          |

## Revision history

| Version | Date       | Author       | Change                                                                                        |
|---------|------------|--------------|-----------------------------------------------------------------------------------------------|
| 0.1     | 2026-09-08 | Laxmi Poudel | Initial draft                                                                                 |
| 0.2     | 2026-09-08 | Laxmi Poudel | CON-7 recorded in the baseline. DEP-5 resolved: embedding model and vector dimension selected |
| 0.3     | 2026-09-08 | Laxmi Poudel | DEP-2 resolved: generation and embedding models selected by measurement                       |
| 0.4     | 2026-09-08 | Laxmi Poudel | Section 3.4.1 points to NOVA-UI-001                                                           |